

MMIS 885: Management Information System

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Computer Science Department, Gombe State University

2025/2026 Session

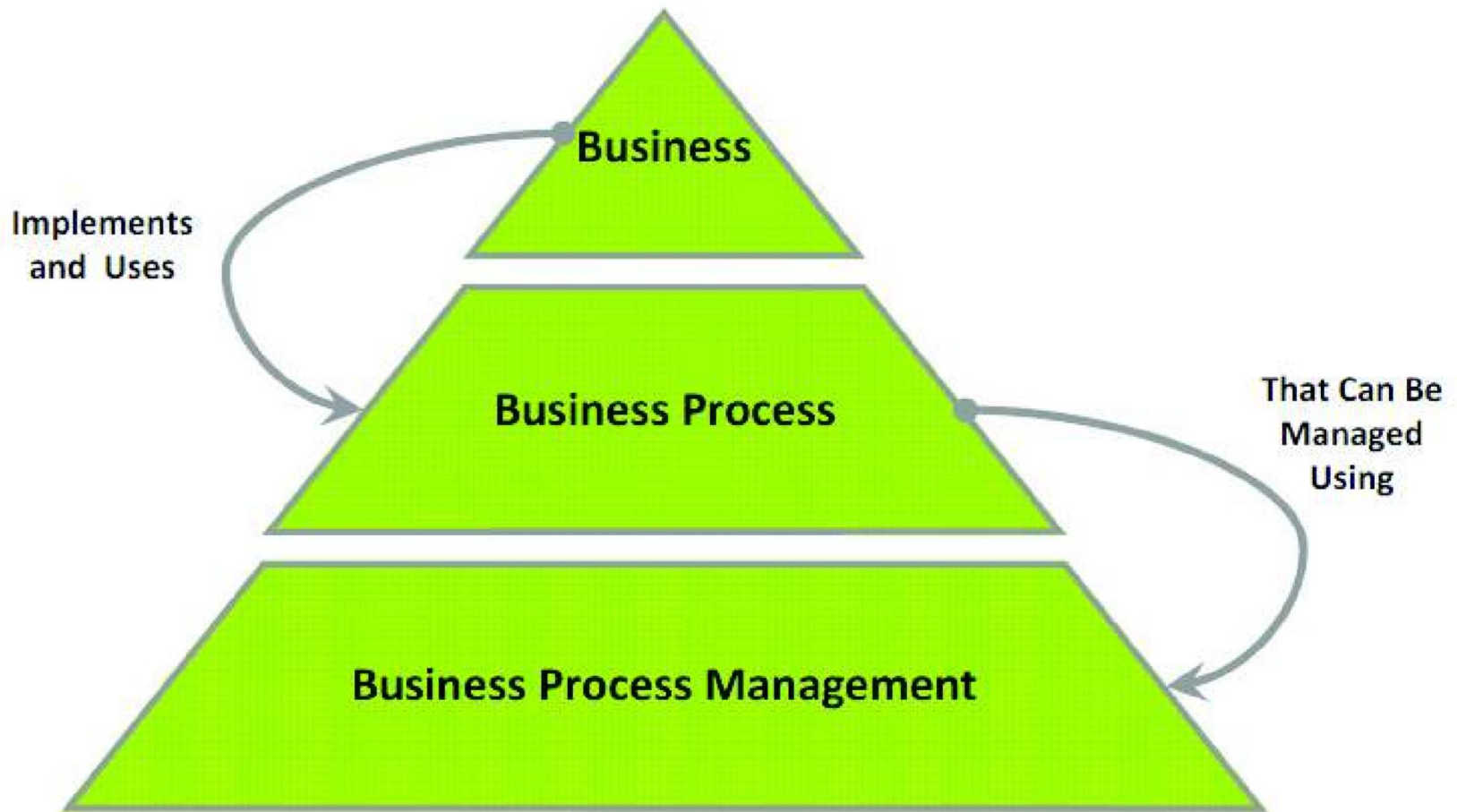
MMIS 885: Lecture 4

Business Process Modelling using BPMN

Business Process Management

- Definitions
- Business
 - Persons or cooperation engaged in commerce, manufacturing, or a service profit seeking enterprise
- Process
 - A systematic series of actions directed to some ends; a continuous set of actions, operations or series of changes taking place in a definite manner
 - A repetitive set of interrelated tasks aimed at achieving a certain goal
- Management
 - The act or manner of managing; handling, direction or control, execution ability; the person or persons controlling the affairs of a business or institution

Business Process Management



Business Process Modelling with Adonis

- Adonis BPM toolkit
 - The Adonis BPM toolkit allows users to model their business processes as well as their working environment
 - The toolkit offers extensive functionality, which allows users to acquire, analyze, simulate, and evaluate business processes and working environments in a cost-effective way

<https://www.adonis-community.com/en/>

The screenshot displays the Adonis:CE website homepage. The browser's address bar shows the URL <https://www.adonis-community.com/en/>. The website's header includes the Adonis:CE logo with the tagline "Your free BPM Tool", a navigation menu with links for HOME, GO-PRO, SUPPORT & FAQs, and LOGIN, a yellow GET STARTED button, and a language selector set to ENGLISH. The main hero section features a background image of two women working on a laptop, with the text "ADONIS:COMMUNITY EDITION" and "YOUR FREE CLOUD BPM TOOL" overlaid. A yellow GET STARTED button is positioned in the center of the hero section. The right sidebar contains a search icon, a list of application icons (including PowerPoint and Google Chrome), and a system tray area showing the desktop environment, language settings (ENG, INTL), and the current time and date (9:48 AM, Sunday, 25/02/2024).

ADONIS:CE
Your free BPM Tool

HOME GO-PRO SUPPORT & FAQs LOGIN GET STARTED ENGLISH

ADONIS:COMMUNITY EDITION
YOUR FREE CLOUD BPM TOOL

GET STARTED

<https://www.adonis-community.com/en/account-registration/>

https://lucid.co/lucidchart

The image shows a web browser window displaying the Lucidchart website. The browser's address bar shows the URL `lucid.co/lucidchart`. The website's navigation bar includes links for Products, Solutions, Resources, Company, and Enterprise, along with buttons for Log in, Contact sales, and Sign up free. Below the navigation bar, the Lucidchart logo is followed by links for Overview, Use cases, and Pricing, and a Watch a demo button.

The main content area features the heading "Diagramming powered by intelligence" and a subheading "Create next-generation diagrams with AI, data, and automation in Lucidchart. Understand and optimize every system and process." Below this is a "Sign up free" button and logos for Google, Microsoft, and Slack. A note at the bottom states: "By registering, you agree to our [Terms of Service](#) and you acknowledge that you have read and understand our [Privacy Policy](#)."

Overlaid on the bottom right of the website is a screenshot of the Lucidchart application interface. The interface shows a "Blank Diagram" workspace with a menu bar (File, Edit, Select, View, Insert, Arrange, Share, Help) and a toolbar. A sidebar on the left contains a "Lucid AI [BETA]" section with a text prompt: "Create a high-level IT infrastructure diagram using Google Cloud networking shapes that show our hybrid environment. Include on-prem servers, cloud services, network components (firewalls, switches, VPN), and key applications connected to user groups. Highlight data flow between systems and include labels for critical systems." Below the prompt, a message reads: "Okay, I've generated the high-level IT diagram you asked for. Would you like to make any edits or add more details to this diagram?" The main workspace displays a generated IT infrastructure diagram with various components like "Google Cloud Service 1", "Application 1", "Network Switch 1", "Firewall 1", "VPN 1", and "User Group 1" connected by lines. A "Have more to add?" input field is at the bottom of the sidebar. An "Activate Windows" watermark is visible in the bottom right corner of the application window.

Sign Up | Free Business Process

adonis-community.com/en/account-registration/

ADONIS:CE

Your free BPM Tool

1 User data

2 Account data

3 Confirmation

E-MAIL *

FORM OF ADDRESS *

munattabra@gmail.com

Ms.

FIRST NAME *

LAST NAME *

Maimuna

Salisu Tabra

I AM A STUDENT *

Yes

No

CONTINUE

ENJOY QUICK & EASY WEB-BASED MODELLING

✓ Modern and intuitive interface

✓ Hover modelling assistant

✓ BPMN 2.0 support, fit for business

Desktop

ENG

INTL

9:46 AM

Sunday

25/02/2024

Sign Up | Free Business Process

adonis-community.com/en/account-registration/#gf_10

ADONIS:CE

Your free BPM Tool

1 User data

2 Account data

3 Confirmation

ORGANIZATION/COMPANY *

Gombe State University

TELEPHONE *

+2347061041226

COUNTRY *

Nigeria

TOOL LANGUAGE *

English

BACK

CONTINUE

PERFORM POWERFUL SEARCH & ANALYSIS

✓ Comprehensive reporting dashboard

✓ Graphical analysis views for all BPM Stakeholders

✓ Various export options

Desktop

9:42 AM
Sunday
25/02/2024

Check Terms of Use and Submit

Sign Up | Free Business Process

adonis-community.com/en/account-registration/#gf_10

ADONIS:CE
Your free BPM Tool

1 User data 2 Account data 3 Confirmation

TERMS OF USE *

☒ By registering for the ADONIS:Community Edition, I agree on the [terms of use](#) and the [privacy policy](#) of BOC Group.

BACK SUBMIT

**NETWORK,
LEARN & GROW
WITH ADONIS:CE**

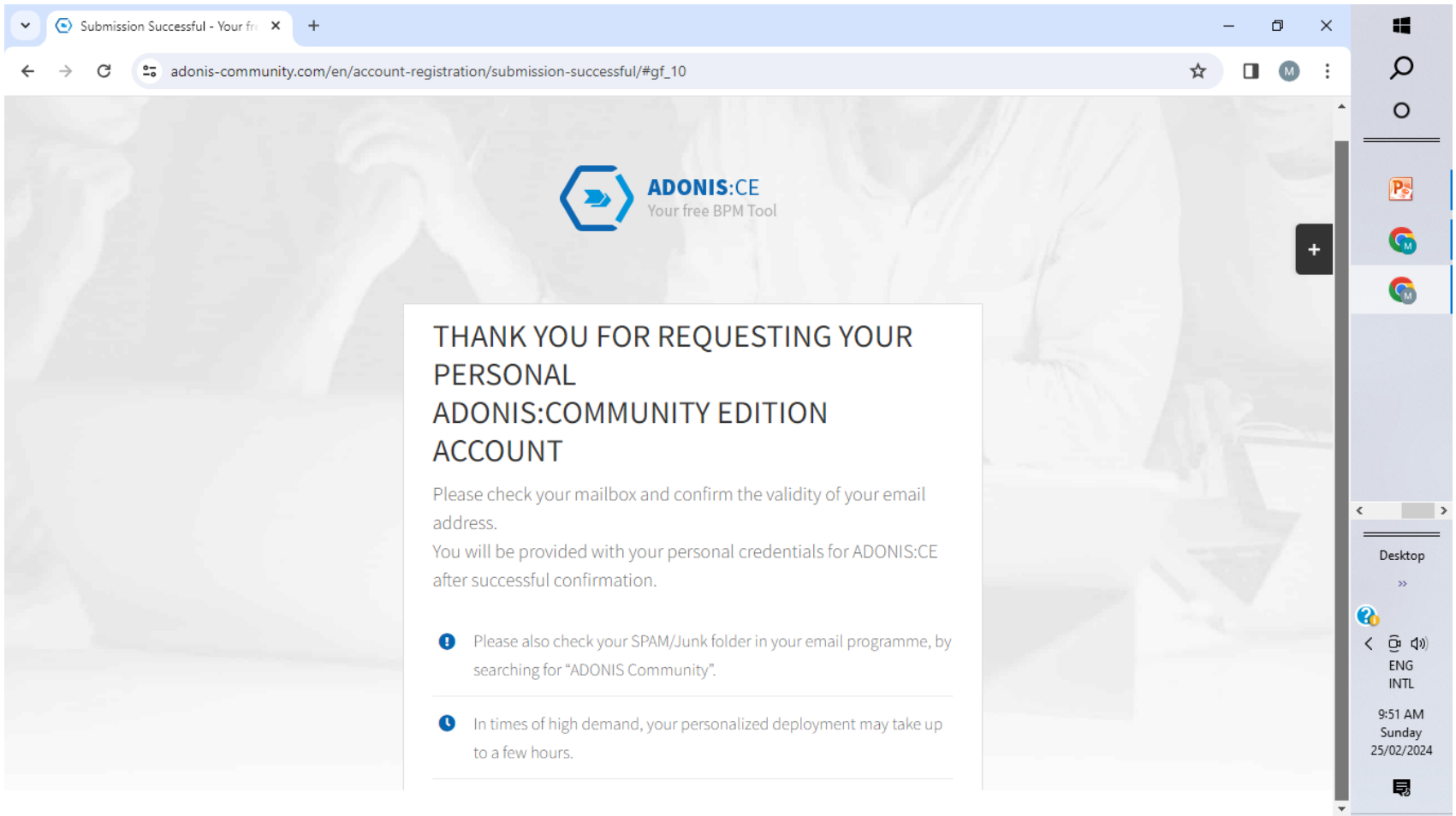
- ✓ Useful sample content
- ✓ Short and comprehensive tutorial videos
- ✓ Specially BPM dedicated expert community

Desktop

ENG
INTL

9:49 AM
Sunday
25/02/2024

Success Report



Submission Successful - Your free BPM Tool

adonis-community.com/en/account-registration/submission-successful/#gf_10

ADONIS:CE
Your free BPM Tool

THANK YOU FOR REQUESTING YOUR PERSONAL ADONIS:COMMUNITY EDITION ACCOUNT

Please check your mailbox and confirm the validity of your email address.
You will be provided with your personal credentials for ADONIS:CE after successful confirmation.

- ! Please also check your SPAM/Junk folder in your email programme, by searching for "ADONIS Community".
- ! In times of high demand, your personalized deployment may take up to a few hours.

Desktop

ENG
INTL

9:51 AM
Sunday
25/02/2024

Verify Email Success Report

The screenshot shows a web browser window with the URL `adonis-community.com/en/account-registration/confirmation-successful/`. The page content is as follows:

THANK YOU FOR CONFIRMING YOUR EMAIL ADDRESS

We are creating your ADONIS:CE account as we speak. You will be provided with your personal account data shortly via email.

Note: In times of high demand, your personalized deployment may take up to a few hours. In such cases, please bear with us! It'll be worth the wait.

In the meantime, explore our ADONIS Marketplace or Knowledge Hub.

BPM Marketplace

Extend. Enhance your ADONIS Portfolio.

Find the perfect addition to meet all your Business Process Management needs.

BPM Knowledge-Library

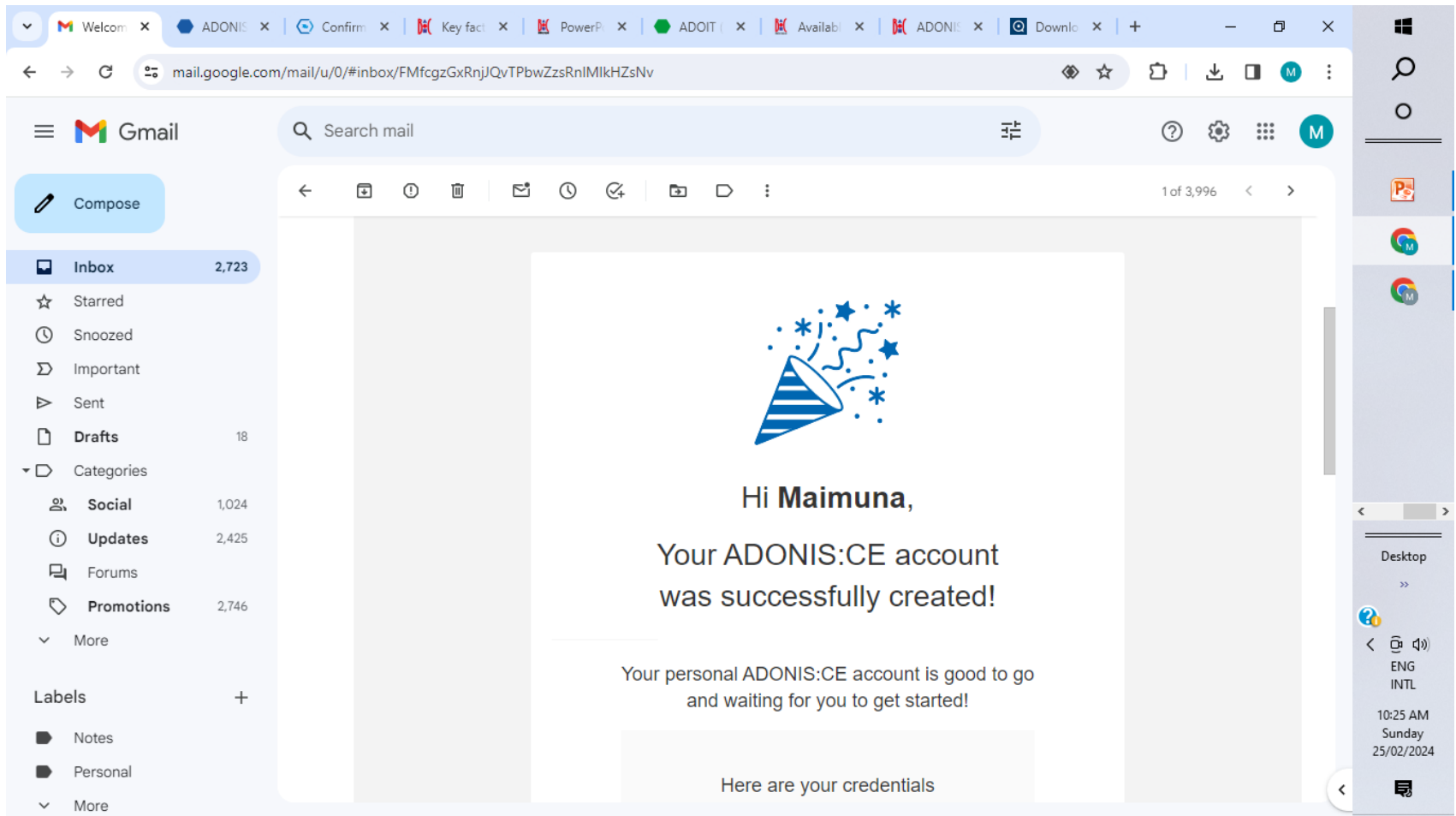
Learn. Gain Knowledge.

Your #1 knowledge platform. Get to know ADONIS better. Grow your BPM skills. Leverage industry expertise and best practices.

KNOWLEDGE HUB

The browser's taskbar on the right shows the Windows Start button, search icon, and task view icon. Below these are icons for PowerPoint, two Chrome browser windows, and a desktop icon. The system tray at the bottom right displays the language as 'ENG INTL', the time as '9:56 AM Sunday 25/02/2024', and a network icon.

Account Created - Message via Email



Create your First Process

The screenshot displays a Gmail interface on a desktop. The browser's address bar shows the email URL: `mail.google.com/mail/u/0/#inbox/FMfcgzGxRnjJQvTPbwZzsRnIMikHZsNv`. The Gmail header includes the 'Gmail' logo, a search bar, and a user profile icon. The left sidebar lists various email folders: 'Compose', 'Inbox' (2,723), 'Starred', 'Snoozed', 'Important', 'Sent', 'Drafts' (18), 'Categories' (Social: 1,024, Updates: 2,425, Forums, Promotions: 2,746), and 'Labels' (Notes, Personal). The main email content area shows a message from ADONIS:CE. The email body contains the text: 'We will shortly provide you with some guidance on your first steps with ADONIS:CE. But for now, don't wait any longer:'. Below this text is a prominent blue button labeled 'CREATE YOUR FIRST PROCESS'. At the bottom of the email, it says 'P.S. You can also log in the tool directly on the [ADONIS:Community website.](#)'. A blue banner at the very bottom of the email encourages following the 'BOC GROUP' on LinkedIn, Twitter, YouTube, and X. The Windows taskbar on the right shows the Start button, search icon, and task view button, with a desktop notification area at the bottom right displaying the time as 10:26 AM on Sunday, 25/02/2024.

Welcom x ADONIS x Confirm x Key fact x PowerPi x ADOIT x Availabl x ADONIS x Downlo x +

mail.google.com/mail/u/0/#inbox/FMfcgzGxRnjJQvTPbwZzsRnIMikHZsNv

Gmail Search mail

Compose

Inbox 2,723

Starred

Snoozed

Important

Sent

Drafts 18

Categories

Social 1,024

Updates 2,425

Forums

Promotions 2,746

More

Labels +

Notes

Personal

More

We will shortly provide you with some guidance on your first steps with ADONIS:CE.

But for now, don't wait any longer:

CREATE YOUR FIRST PROCESS

P.S. You can also log in the tool directly on the [ADONIS:Community website.](#)

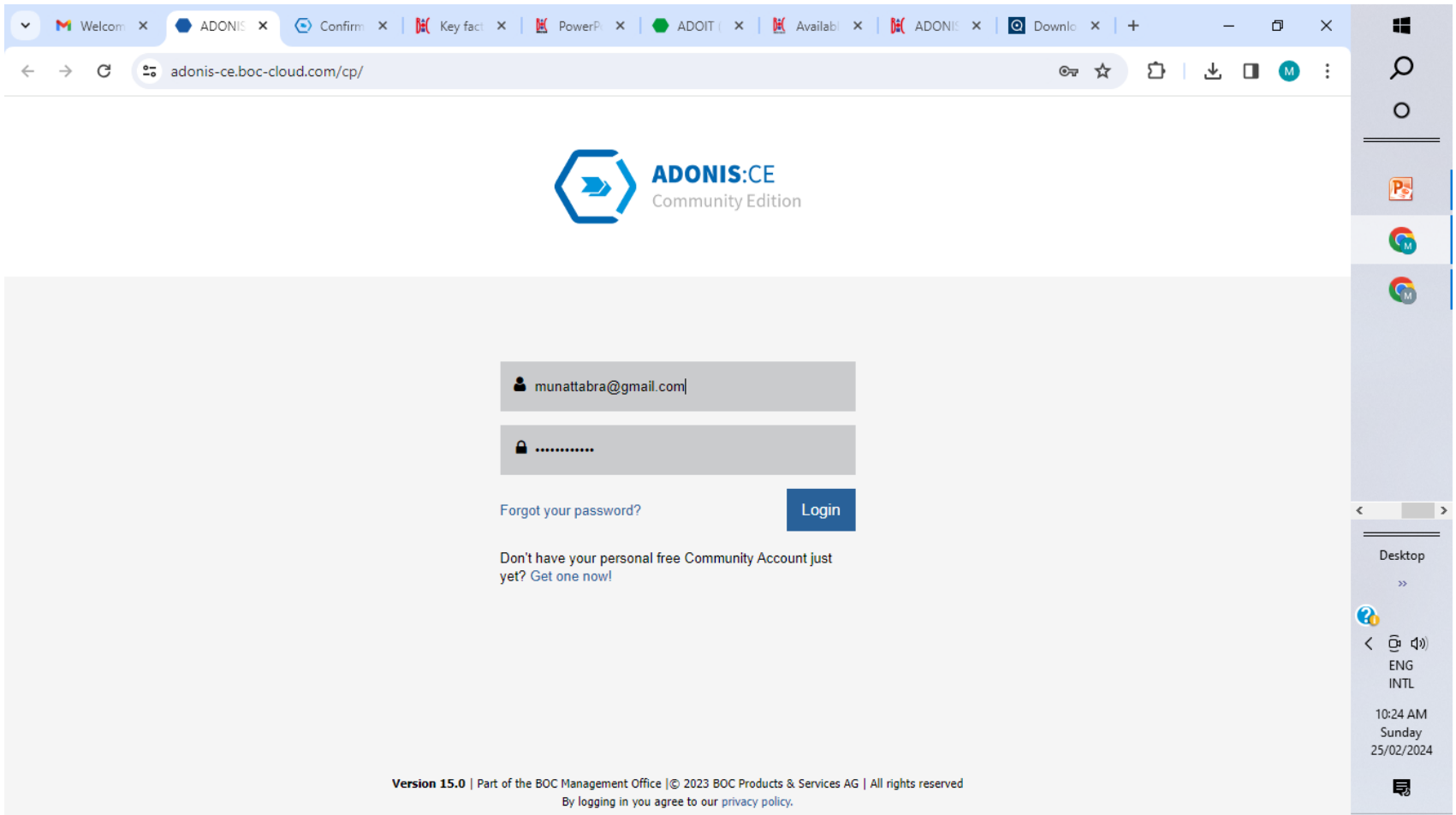
FOLLOW BOC GROUP

in tw YouTube X

Desktop

10:26 AM Sunday 25/02/2024

Login with the provided Credentials



The screenshot shows a web browser window with the URL `adonis-ce.boc-cloud.com/cp/`. The page features the ADONIS:CE Community Edition logo at the top. Below the logo, there is a login form with two input fields: the first contains the email address `munattabra@gmail.com`, and the second contains a masked password. To the right of the password field is a blue "Login" button. Below the login fields, there is a link for "Forgot your password?". At the bottom of the page, a footer contains the text: "Version 15.0 | Part of the BOC Management Office | © 2023 BOC Products & Services AG | All rights reserved. By logging in you agree to our [privacy policy](#)."

ADONIS:CE
Community Edition

`munattabra@gmail.com`

.....

[Forgot your password?](#) [Login](#)

Don't have your personal free Community Account just yet? [Get one now!](#)

Version 15.0 | Part of the BOC Management Office | © 2023 BOC Products & Services AG | All rights reserved.
By logging in you agree to our [privacy policy](#).

ADONIS User Interface

The screenshot displays the ADONIS User Interface within a web browser. The browser's address bar shows the URL `adonis-ce.boc-cloud.com/main.view#0`. The application's top navigation bar includes a home icon, a dropdown menu labeled "Design & Document", and buttons for "+ New", "Find", "Analyse", "Models: 61 / 256", and "Objects: 605 / 2000". On the right side of the navigation bar are links for "Give feedback", "Help", and a user profile icon.

The main content area features the ADONIS:CE logo with the tagline "Your free BPM Tool" on the left. On the right, there is a promotional banner for the ADONIS mobile app, which includes a QR code and the text "GET THE ADONIS MOBILE APP TODAY! BPM on the Go". Below the banner, there are two tabs: "Recently Opened" (which is active) and "Favourites". The "Recently Opened" tab shows a clock icon and the message "Nothing to see yet. Your recently opened models will appear here."

The right sidebar of the browser contains several icons, including a search icon, a home icon, and a "Desktop" button. At the bottom of the sidebar, there is a language selector showing "ENG" and "INTL", and a timestamp indicating "11:36 AM Sunday 25/02/2024".

- Link and Login credentials will be provided for accessing the ADONIS Community Guide.
- Follow guide to the Course catalog and take courses to improve your business process management skills.

ENGLISH → ADONIS → Free ADONIS Fundamentals

ADONIS Fundamentals 1: Business Process Diagrams

(ADONIS_F01_ENG)

Focus: Process Modelling Fundamentals for Business Process Diagrams

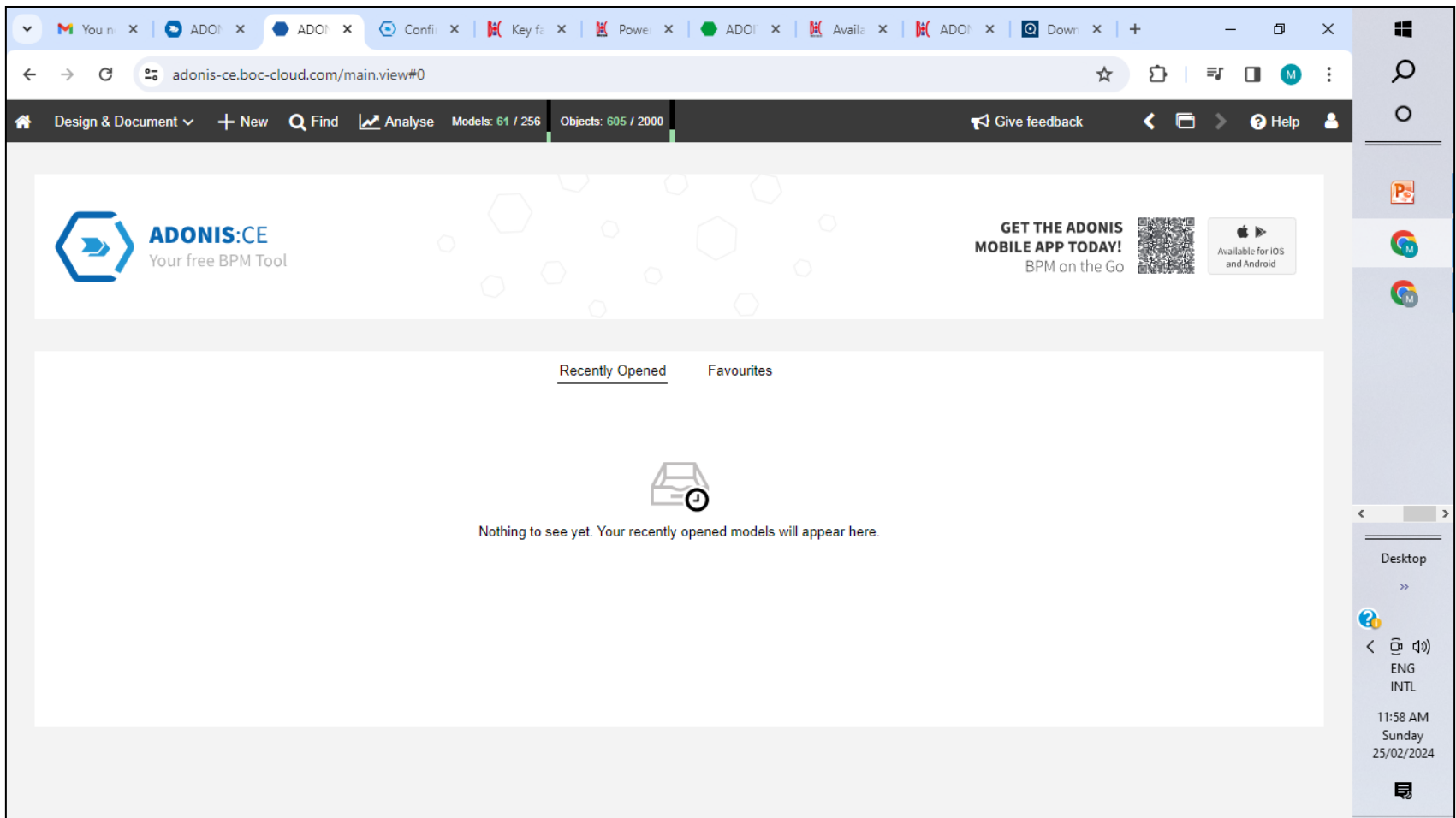
This course provides you with the fundamentals on Process Modelling. It will help you understand how to capture individual processes in your organization, make them clear for everyone and uncover inefficiencies.

You will learn:

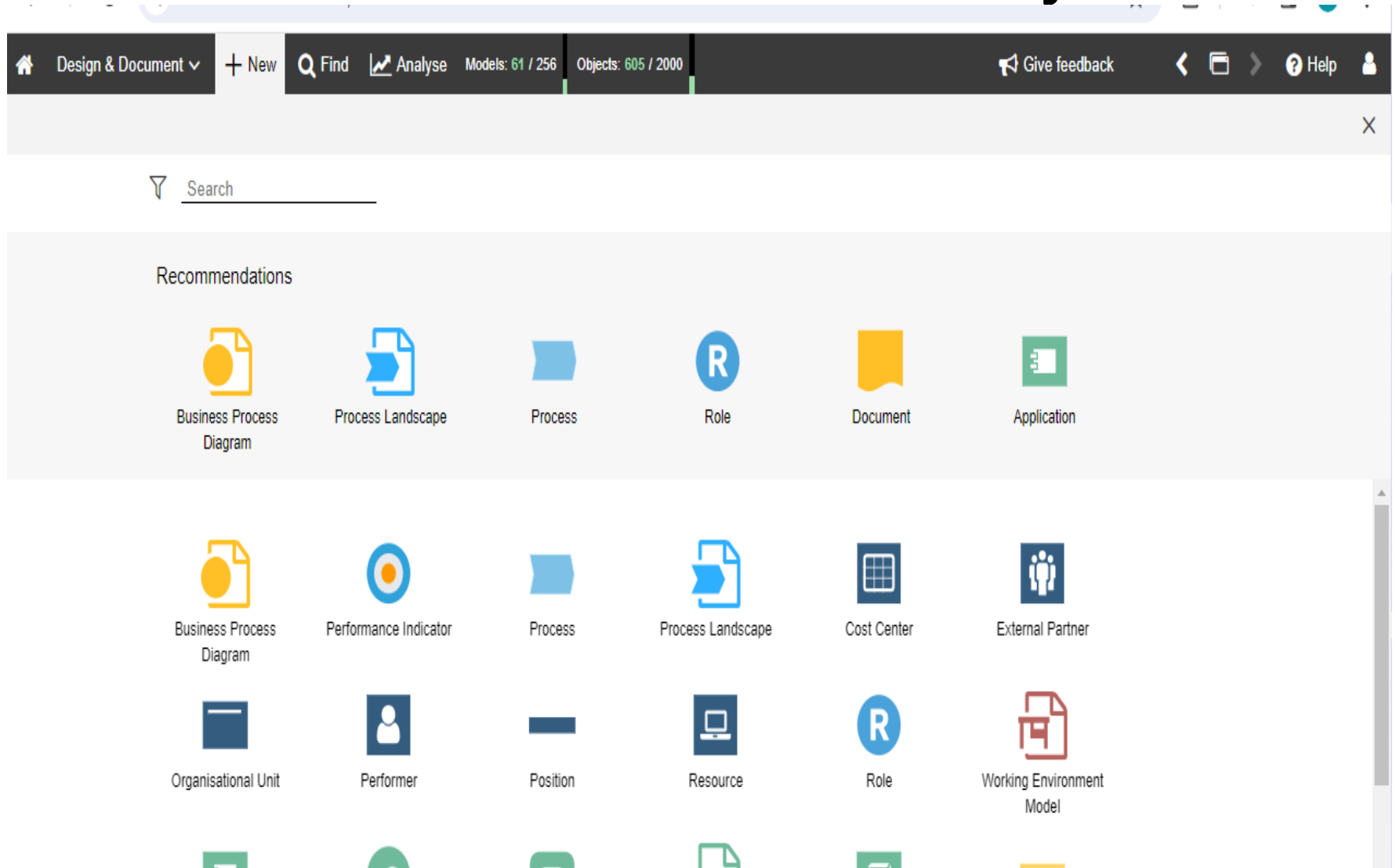
- How to navigate the ADONIS User Interface
- How to create a Business Process Diagram
- How to add responsibilities and alternative paths
- How to enrich your Business Process Diagram
- How to Export/Share your results

Log into your ADONIS account and follow along!

Explore ADONIS User Interface



Click “New” Tab to create new content- Models and Objects



How to create a Business Process Diagram

Business Process Diagrams in ADONIS help you depict a sequence of activities needed to achieve a specific result. They can portray a variety of things, such as:

- 1 How to manage travel expenses
- 2 How your IT department resolves a problem
- 3 How to handle HR requests
- 4 And many more

ADONIS uses the **BPMN 2.0** modeling language – the global standard when it comes to visualizing process flows.

In this tutorial, we'll model the **happy path** (also known as **happy flow**) of an ideal execution of a pizza delivery process. The happy path refers to a case where the process is completed without running into any exceptions.

But before we begin, let's grasp the key elements of process modelling.

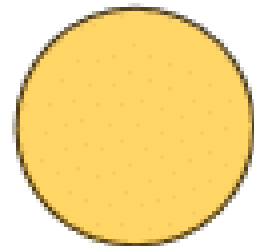
Elements of Process Modeling

- Start Event
- End Event
- Sequence Flow
- Task (Activity)
- Gateways
- Pools and lane

Start Event

A **Start Event** *initiates a process*, denoted by a circular shape with a thin outline and an open centre (leaving room for additional markers)

- Start Events can be triggered by a variety of things, such as:
- Manual actions, e.g., user button clicks
- Messages from external systems
- Timer expiration
- Specific condition occurrences



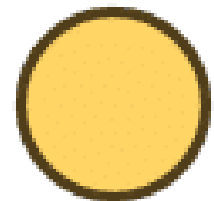
Start Event

End Event

End Events mark the *ending of a process*, depicted as a circled shape with a bold outline and an open centre (for additional markers to be added).

End Events are triggered by:

- Successful completion of all process activities
- Satisfaction of specified conditions
- Process cancellation



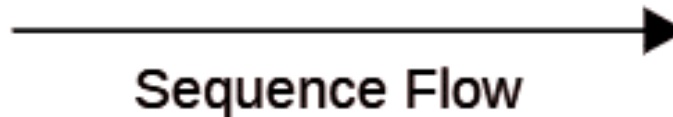
End Event

Sequence Flow

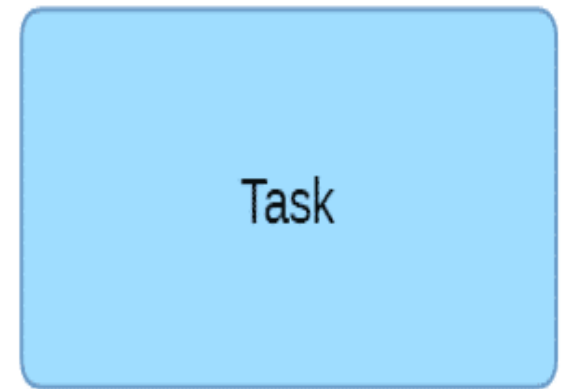
Sequence Flows depict the ordered sequence of activities in a process.

They show the order in which tasks, events, and decisions happened

They must not intersect with Pool boundaries and are illustrated as solid lines with arrows.



Task (Activity)



Tasks (Activities)

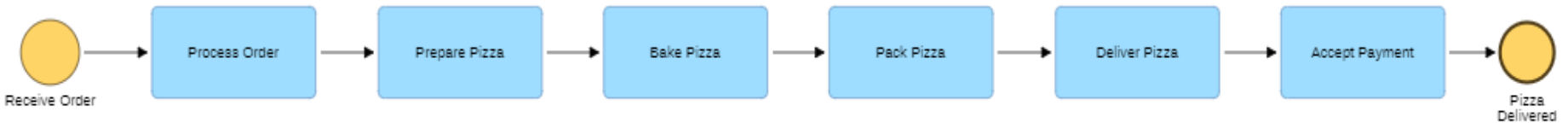
Represents a task performed in a process

Tasks or activities are usually performed by a person, a system, or another process

Tasks are central to business process modelling, representing the work performed by an organization within its processes. They encompass a range of actions and operations aimed at achieving specific objectives within the workflow.

Create Business Process diagram

The happy path of a pizza delivery Process Model



How to add responsibilities and alternative paths

For the sake of the example, let us acknowledge the fact, that not every order can be processed, e.g. for the lack of a certain ingredient, that was ordered.

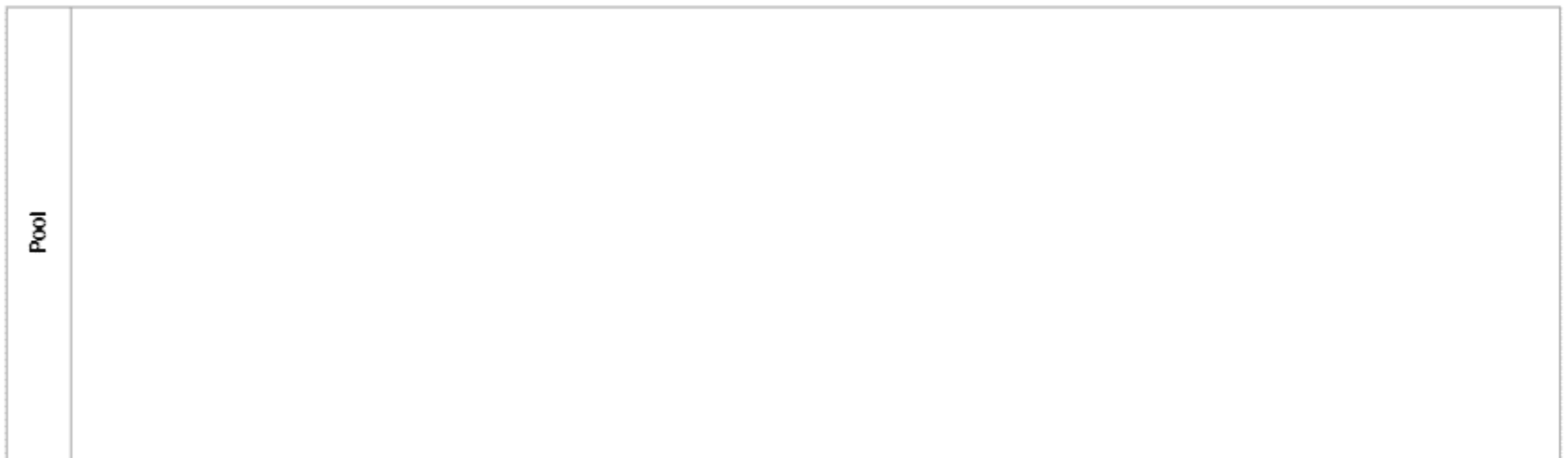
To depict the different responsibilities and paths a process can take, we will look into the creation of **Pools and Lanes**, as well as into modeling of **alternative paths**.

Showing responsibilities with Pools and Lanes

- Since various steps of your processes are carried out by different people or systems, it is now also the best time to graphically depict this through the use of **Pools and Lanes** in ADONIS.
- By incorporating **Pools and Lanes** at this stage, you will ease further modelling and create transparency on the nature of the workflow for all readers, from the start.

Pool

- A **Pool** is like a big container that represents the overall process you are modelling. It is depicted as a rectangle that encompasses all the Tasks and Lanes associated with that business process



Lane

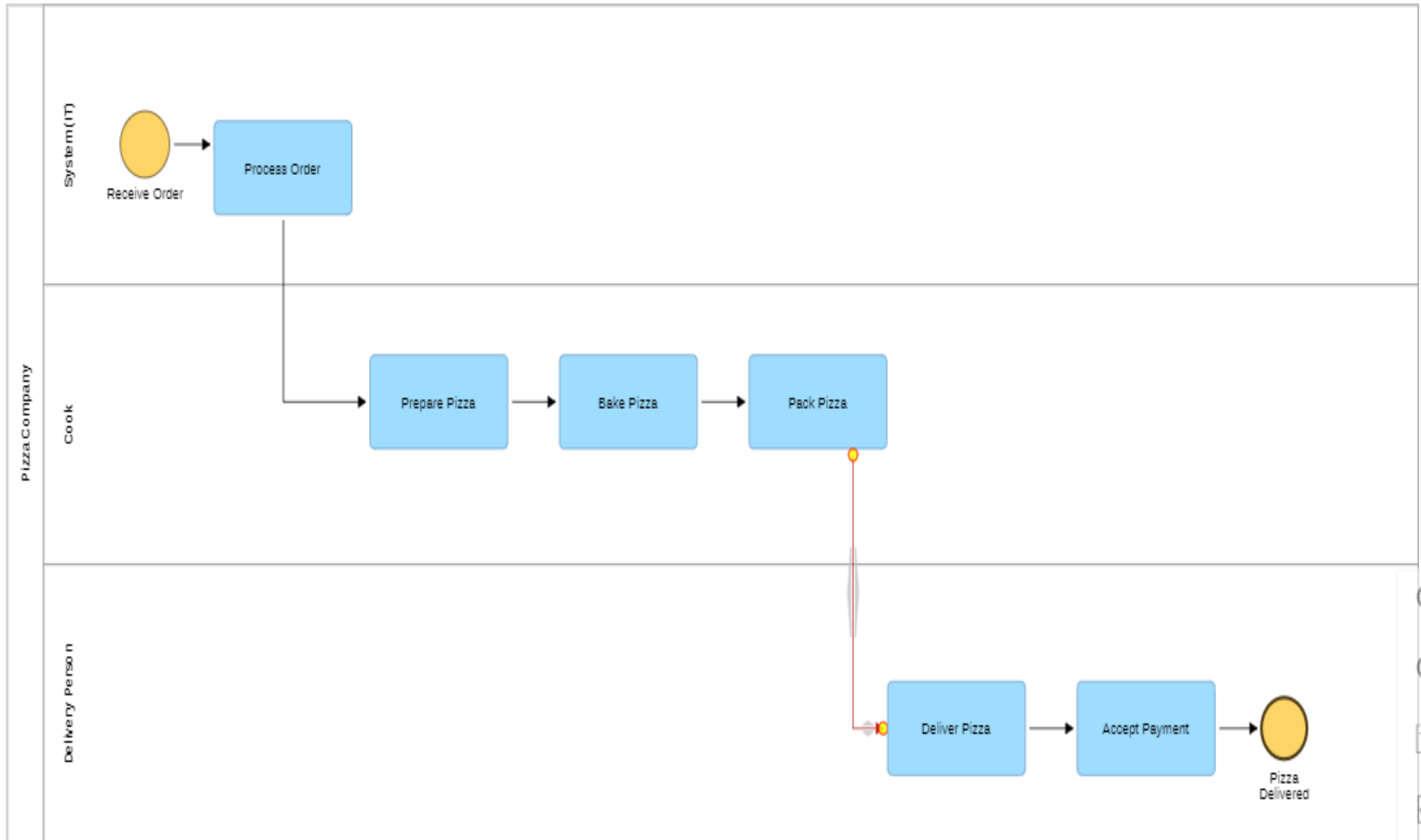
- Within a Pool, **Lanes** function as rows that categorize and organize the tasks based on who is doing them and what job they have. The Lanes are visualized as vertical sections within the Pool, typically named after their associated role or department. Such naming makes it easier to tell which tasks belong to different groups within the same Pool, making the diagram easier to understand and read.

Lane

Lane

- We will now add a Pool (Pizza Company) and Lanes - System (IT), Cook and Delivery Person - to our example.

Process Diagram with Pool and Lane

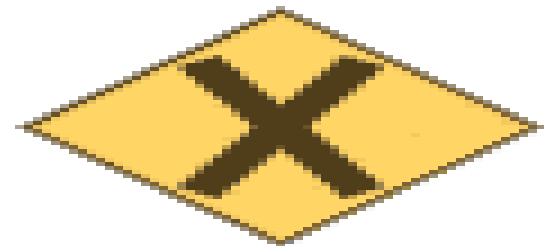


Modelling decisions and alternative/parallel paths

In most cases, you also have to consider alternative process paths – meaning specific cases in which the process comes to a different end result, or where other tasks need to be performed for a specific reason. To visualize this, we use **Gateways** in ADONIS. In the following, we'll cover the most common Gateway types.

Exclusive Gateway (XOR)

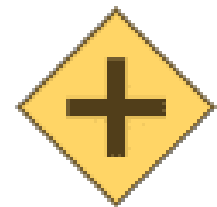
- This type of gateway allows only one of the outgoing connectors to follow. It comes with a condition that must be met for one of the outgoing connectors to become active. It can be used to make a decision, like approving or rejecting a request.



Exclusive
Gateway

Parallel Gateway (AND)

- This type of gateway allows all of the outgoing connectors. It doesn't have a specific condition, therefore all outgoing connectors can be used when it is activated. As the name implies, this gateway can execute multiple activities in parallel, such as when several people need to approve a request.

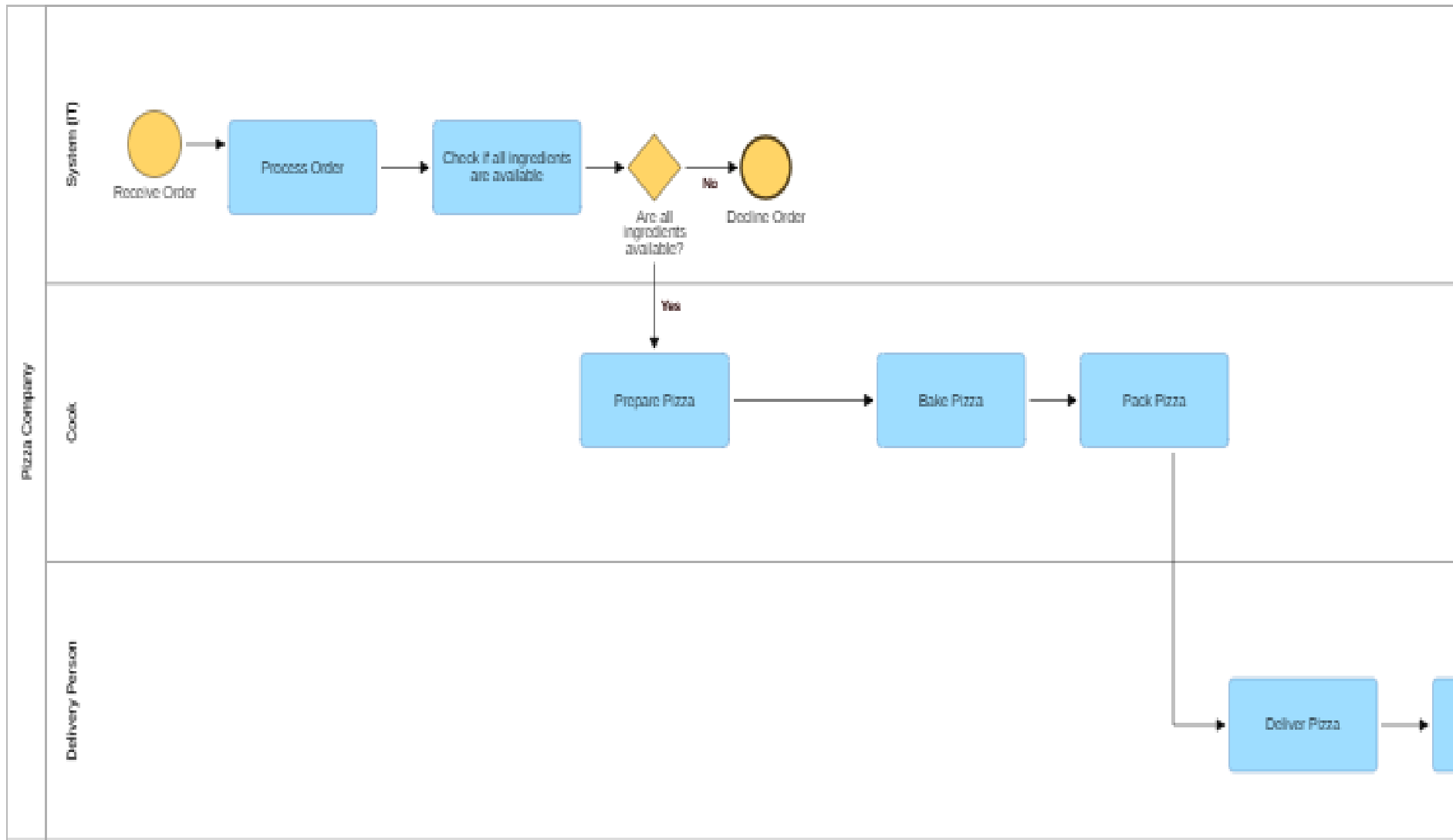


Parallel
Gateway

Using the Exclusive Gateways

- In our **Pizza Delivery** process, we encounter situations where the pizza company doesn't have all the ingredients to complete all orders. Therefore, it's important to check if everything is available and, if needed, decline the order. This is where the **Exclusive Gateways** come into play.

- An Exclusive Gateway needs to be labelled with a simple question – ideally one that can be answered with Yes/No. The arrows pointing away from it need to be clearly labelled too (e.g. with Yes or No). Sometimes a Closing Gateway is also used to merge the incoming flows – however, in ADONIS, this is not necessary. To ensure that the question in the gateway can actually be answered, you can insert a so-called check activity before the gateway.



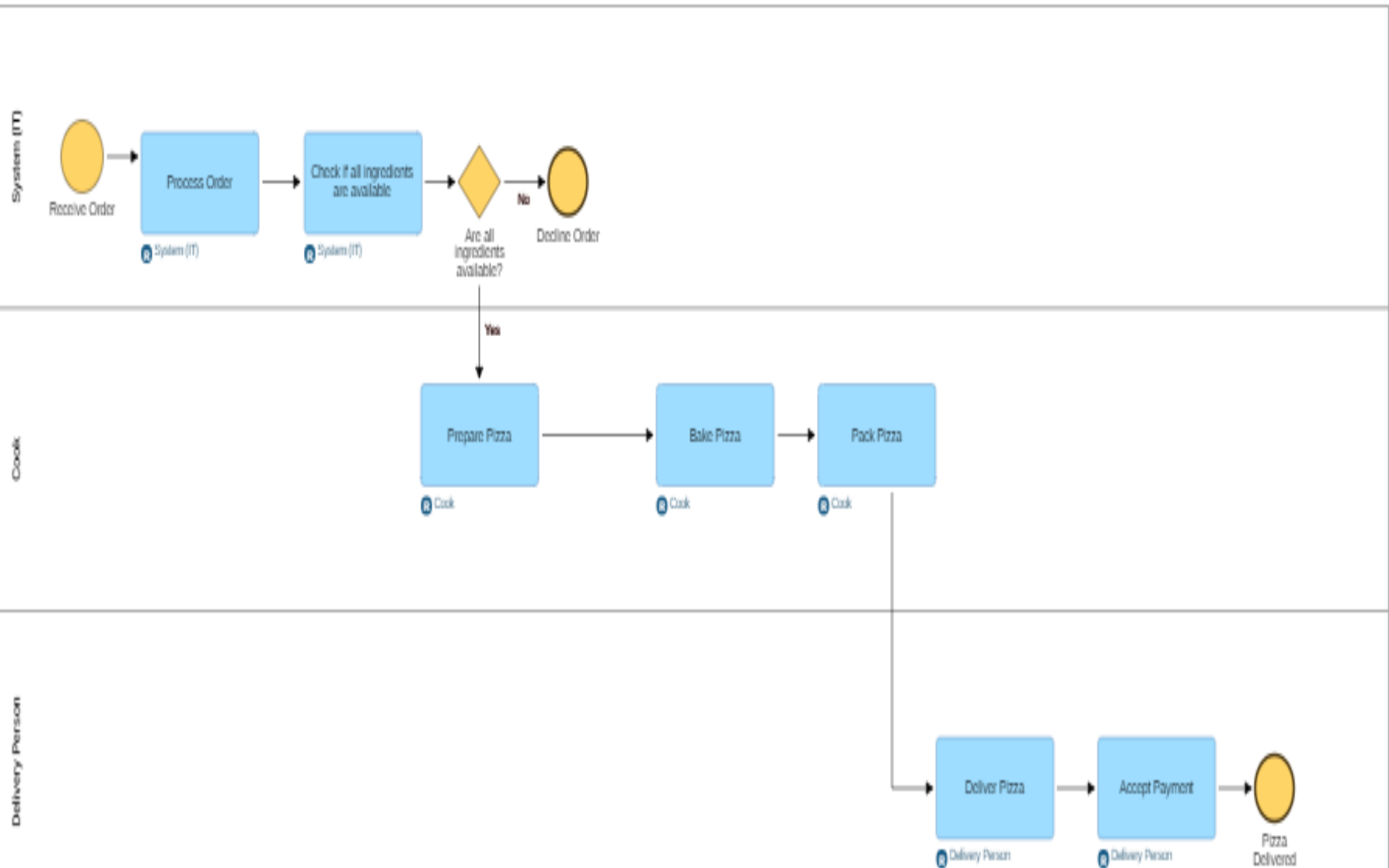
How to enrich your Business Process Diagram

- A process diagram typically also should provide more information than the flow of activities. This starts from simple descriptive texts for each task, and can range to highlighting who is carrying out certain tasks or which equipment is needed to complete the job.
- For that reason, you can link various assets to your process, such as documents, roles, input/output and more.
- First of all, we will assign **roles** to the tasks they are responsible for.
- However, only the role "System (IT)" should already exist in your repository. Therefore, we need to also create new roles for the other two lanes: "Cook" and "Delivery Person".

Tasks and corresponding Roles

Task	Responsible for execution
Process order	System (IT)
Check if all ingredients are available	System (IT)
Decline order	System (IT)
Prepare pizza	Cook
Bake pizza	Cook
Pack pizza	Cook
Deliver pizza	Delivery Person
Collect payment	Delivery Person

Process Diagram with Roles attached



How to Export/Share your results

ADONIS provides you with options to share your process models. The need to share can have many different reasons, as for example:

- sharing a draft so you can get feedback from colleagues or distinct process experts to see if you got it right,
- providing the process diagram as input on a specific discussion or workshop,
- and printing the diagram and hanging it up the wall, so you and your colleagues will always know how to execute said process.
- So, let's have a look at how you can get your work results to others.

Option 1: Print PDF or Export image

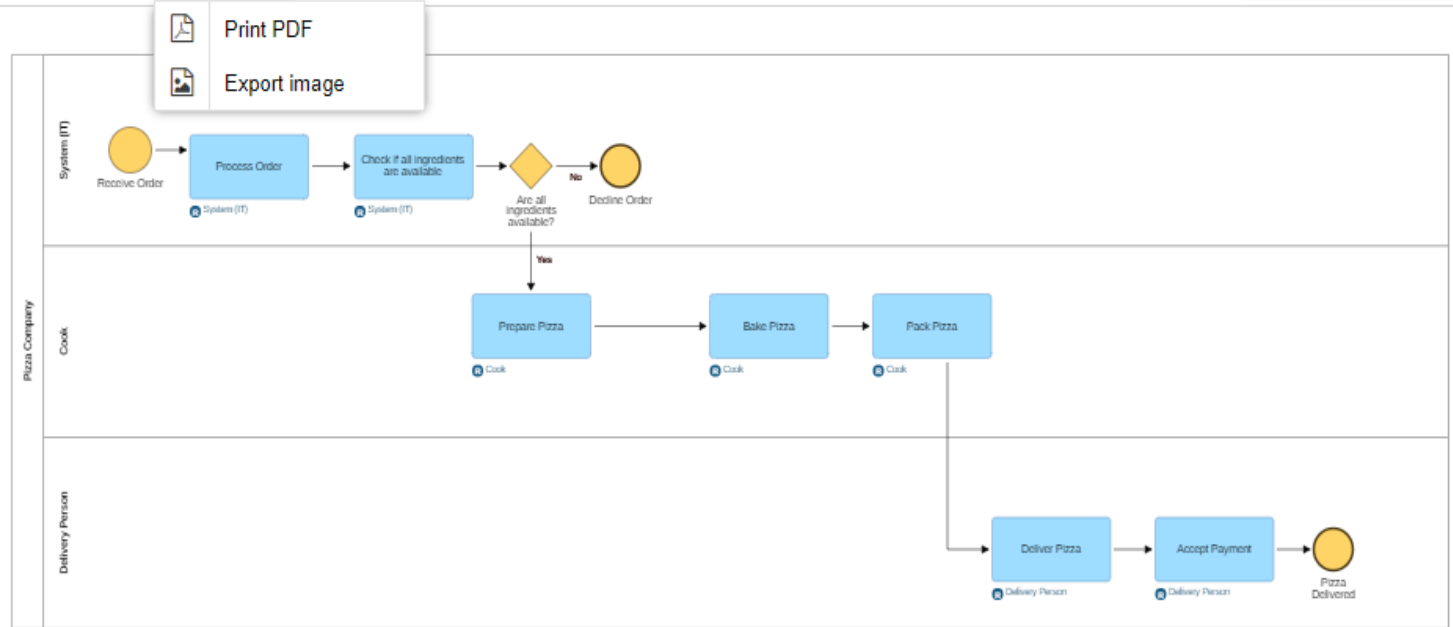
Process diagrams and all other sorts of graphical models can be exported as a PDF document. You can save this document on your computer, share it with colleagues or print it on paper.

- Click the Export button in the menu bar of the open model.
- Choose "Print PDF" or "Export image", depending on what your needs are. This action opens the print dialogue.
- Adjust your preferred print settings.
- Confirm by selecting "Print."

📄 Pizza Business Process Diagram 1 ☆ ⌵

Tool palette containing various BPMN symbols and connectors:

- Selection tool (mouse cursor)
- Zoom in/out tools
- Flow connector (arrow)
- Start event (yellow circle)
- Intermediate event (blue circle)
- End event (yellow circle)
- Activity (rounded rectangle)
- Exclusive gateway (yellow diamond)
- Inclusive gateway (yellow circle)
- Split/Join gateway (yellow diamond with +)
- Message boundary (dashed line)
- Message start/end (yellow rectangle)



Option 2: Create a pre-defined report

- To provide a more detailed export of your process diagram, that also contains additional data from the tasks' properties, you can create various pre-defined reports in PDF or editable formats. The most typical case at this stage will be a "Model report". You can trigger it by:
- Opening the model you want to create a report on.
- Clicking on the **More menu (...)** in the top right corner of the model.
- Pointing to **Reports**, and then choose "**Model report..**". The settings dialogue opens.
- Adjusting the settings to your needs and click on "**OK**".
- Your report will be downloaded to your device!

Review Questions

- 1. Define the following terms as used in the Adonis or Lucidchart Business Process Modelling tool:
 - Start Event
 - Task (Activity)
 - End Event
 - Sequence Flow
 - Pool
 - Lane
 - Exclusive Gateway.
- 2. Think of any business Process in your organization and draw a Business Process Model Diagram in Adonis or Lucidchart.

Presentation Assignments

Due Date: 25/07/2026

S/N	Topics	Group
1	Enterprise systems - concepts, trends and challenges	Group 2
2	Ecommerce and Digital Business	Group 1
3	Information system project Management	Group 4
4	Information Security Management in Organizations	Group 3